

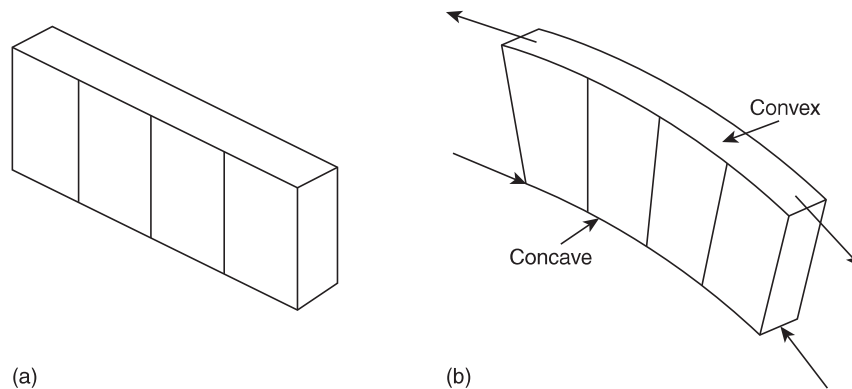


Fig. 16.1 Bending of a rubber eraser.

Now rotate the eraser so that its shorter sides are vertical and apply the same pressure with your fingers. The eraser again bends but now requires much less effort. It follows that the geometry and orientation of a beam section must affect its *bending stiffness*. This is more readily demonstrated with a plastic ruler. When flat it requires hardly any effort to bend it but when held with its width vertical it becomes almost impossible to bend.

16.1 Symmetrical bending

Although symmetrical bending is a special case of the bending of beams of arbitrary cross-section, we shall investigate the former first, so that the more complex general case may be more easily understood.

Symmetrical bending arises in beams which have either singly or doubly symmetrical cross-sections; examples of both types are shown in Fig. 16.2.

Suppose that a length of beam, of rectangular cross-section, say, is subjected to a pure, sagging bending moment, M , applied in a vertical plane. We shall define this later as a negative bending moment. The length of beam will bend into the shape shown in Fig. 16.3(a) in which the upper surface is concave and the lower convex. It can be seen that the upper longitudinal fibres of the beam are compressed while the lower fibres are stretched. It follows that, as in the case of the eraser, between these two extremes there are fibres that remain unchanged in length.

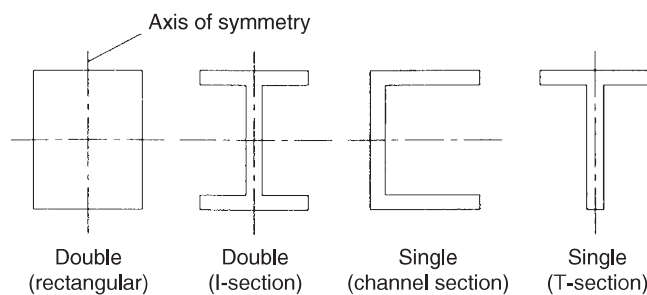


Fig. 16.2 Symmetrical section beams.

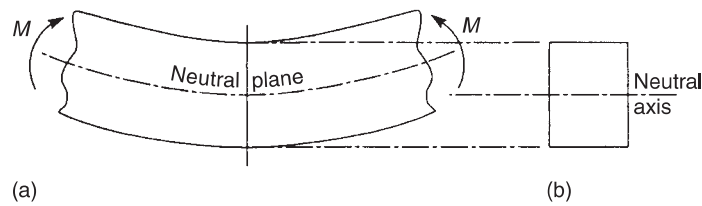


Fig. 16.3 Beam subjected to a pure sagging bending moment.

The direct stress therefore varies through the depth of the beam from compression in the upper fibres to tension in the lower. Clearly the direct stress is zero for the fibres that do not change in length; we have called the plane containing these fibres the *neutral plane*. The line of intersection of the neutral plane and any cross-section of the beam is termed the *neutral axis* (Fig. 16.3(b)).

The problem, therefore, is to determine the variation of direct stress through the depth of the beam, the values of the stresses and subsequently to find the corresponding beam deflection.

16.1.1 Assumptions

The primary assumption made in determining the direct stress distribution produced by pure bending is that plane cross-sections of the beam remain plane and normal to the longitudinal fibres of the beam after bending. Again, we saw this from the lines on the side of the eraser. We shall also assume that the material of the beam is linearly elastic, i.e. it obeys Hooke's law, and that the material of the beam is homogeneous.

16.1.2 Direct stress distribution

Consider a length of beam (Fig. 16.4(a)) that is subjected to a pure, sagging bending moment, M , applied in a vertical plane; the beam cross-section has a vertical axis of symmetry as shown in Fig. 16.4(b). The bending moment will cause the length of beam

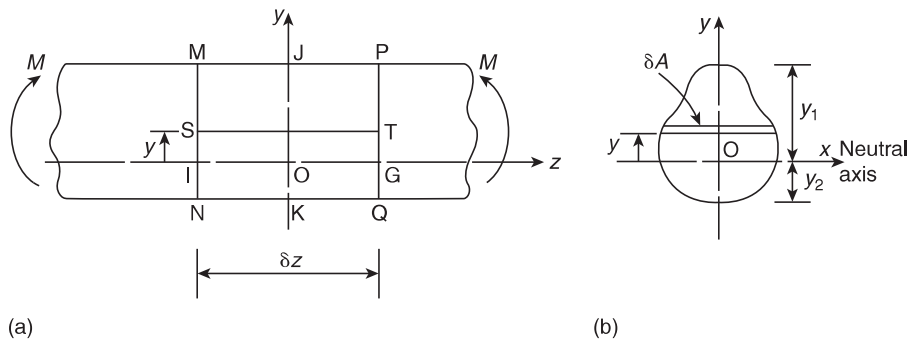


Fig. 16.4 Bending of a symmetrical section beam.

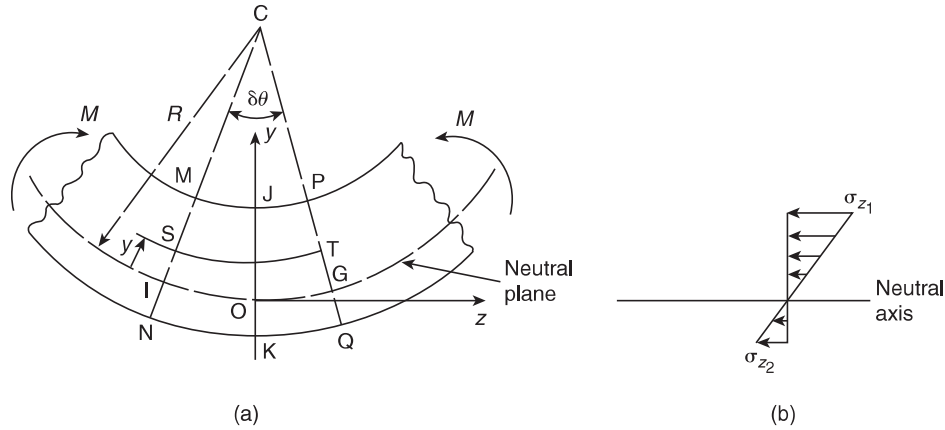


Fig. 16.5 Length of beam subjected to a pure bending moment.

to bend in a similar manner to that shown in Fig. 16.3(a) so that a neutral plane will exist which is, as yet, unknown distances y_1 and y_2 from the top and bottom of the beam, respectively. Coordinates of all points in the beam are referred to axes $Oxyz$ in which the origin O lies in the neutral plane of the beam. We shall now investigate the behaviour of an elemental length, δz , of the beam formed by parallel sections MIN and PGQ (Fig. 16.4(a)) and also the fibre ST of cross-sectional area δA a distance y above the neutral plane. Clearly, before bending takes place $MP = IG = ST = NQ = \delta z$.

The bending moment M causes the length of beam to bend about a *centre of curvature* C as shown in Fig. 16.5(a). Since the element is small in length and a pure moment is applied we can take the curved shape of the beam to be circular with a *radius of curvature* R measured to the neutral plane. This is a useful reference point since, as we have seen, strains and stresses are zero in the neutral plane.

The previously parallel plane sections MIN and PGQ remain plane as we have demonstrated but are now inclined at an angle $\delta\theta$ to each other. The length MP is now shorter than δz as is ST while NQ is longer; IG , being in the neutral plane, is still of length δz . Since the fibre ST has changed in length it has suffered a strain ε_z which is given by

$$\varepsilon_z = \frac{\text{change in length}}{\text{original length}}$$

Then

$$\varepsilon_z = \frac{(R - y)\delta\theta - \delta z}{\delta z}$$

i.e.

$$\varepsilon_z = \frac{(R - y)\delta\theta - R\delta\theta}{R\delta\theta}$$

so that

$$\varepsilon_z = -\frac{y}{R} \quad (16.1)$$

The negative sign in Eq. (16.1) indicates that fibres in the region where y is positive will shorten when the bending moment is negative. Then, from Eq. (1.40), the direct stress σ_z in the fibre ST is given by

$$\sigma_z = -E \frac{y}{R} \quad (16.2)$$

The direct or normal force on the cross-section of the fibre ST is $\sigma_z \delta A$. However, since the direct stress in the beam section is due to a pure bending moment, in other words there is no axial load, the resultant normal force on the complete cross-section of the beam must be zero. Then

$$\int_A \sigma_z \, dA = 0 \quad (16.3)$$

where A is the area of the beam cross-section.

Substituting for σ_z in Eq. (16.3) from (16.2) gives

$$-\frac{E}{R} \int_A y \, dA = 0 \quad (16.4)$$

in which both E and R are constants for a beam of a given material subjected to a given bending moment. Therefore

$$\int_A y \, dA = 0 \quad (16.5)$$

Equation (16.5) states that the first moment of the area of the cross-section of the beam with respect to the neutral axis, i.e. the x axis, is equal to zero. Thus we see that *the neutral axis passes through the centroid of area of the cross-section*. Since the y axis in this case is also an axis of symmetry, it must also pass through the centroid of the cross-section. Hence the origin, O, of the coordinate axes, coincides with the centroid of area of the cross-section.

Equation (16.2) shows that for a sagging (i.e. negative) bending moment the direct stress in the beam section is negative (i.e. compressive) when y is positive and positive (i.e. tensile) when y is negative.

Consider now the elemental strip δA in Fig. 16.4(b); this is, in fact, the cross-section of the fibre ST. The strip is above the neutral axis so that there will be a *compressive* force acting on its cross-section of $\sigma_z \delta A$ which is *numerically* equal to $(Ey/R)\delta A$ from Eq. (16.2). Note that this force will act at all sections along the length of ST. At S this force will exert a clockwise moment $(Ey/R)y\delta A$ about the neutral axis while at T the force will exert an identical anticlockwise moment about the neutral axis. Considering either end of ST we see that the moment resultant about the neutral axis of the stresses on all such fibres must be *equivalent* to the applied negative moment M , i.e.

$$M = - \int_A E \frac{y^2}{R} \, dA$$

or

$$M = -\frac{E}{R} \int_A y^2 \, dA \quad (16.6)$$

The term $\int_A y^2 dA$ is known as the *second moment of area* of the cross-section of the beam about the neutral axis and is given the symbol I . Rewriting Eq. (16.6) we have

$$M = -\frac{EI}{R} \quad (16.7)$$

or, combining this expression with Eq. (16.2)

$$\frac{M}{I} = -\frac{E}{R} = \frac{\sigma_z}{y} \quad (16.8)$$

From Eq. (16.8) we see that

$$\sigma_z = \frac{My}{I} \quad (16.9)$$

The direct stress, σ_z , at any point in the cross-section of a beam is therefore directly proportional to the distance of the point from the neutral axis and so varies linearly through the depth of the beam as shown, for the section JK, in Fig. 16.5(b). Clearly, for a positive bending moment σ_z is positive, i.e. tensile, when y is positive and compressive (i.e. negative) when y is negative. Thus in Fig. 16.5(b)

$$\sigma_{z,1} = \frac{My_1}{I} \text{ (compression)} \quad \sigma_{z,2} = \frac{My_2}{I} \text{ (tension)} \quad (16.10)$$

Furthermore, we see from Eq. (16.7) that the curvature, $1/R$, of the beam is given by

$$\frac{1}{R} = \frac{M}{EI} \quad (16.11)$$

and is therefore directly proportional to the applied bending moment and inversely proportional to the product EI which is known as the *flexural rigidity* of the beam.

Example 16.1

The cross-section of a beam has the dimensions shown in Fig. 16.6(a). If the beam is subjected to a negative bending moment of 100 kN m applied in a vertical plane, determine the distribution of direct stress through the depth of the section.

The cross-section of the beam is doubly symmetrical so that the centroid, C, of the section, and therefore the origin of axes, coincides with the mid-point of the web. Furthermore, the bending moment is applied to the beam section in a vertical plane so that the x axis becomes the neutral axis of the beam section; we therefore need to calculate the second moment of area, I_{xx} , about this axis.

$$I_{xx} = \frac{200 \times 300^3}{12} - \frac{175 \times 260^3}{12} = 193.7 \times 10^6 \text{ mm}^4 \text{ (see Section 16.4)}$$

From Eq. (16.9) the distribution of direct stress, σ_z , is given by

$$\sigma_z = -\frac{100 \times 10^6}{193.7 \times 10^6} y = -0.52y \quad (i)$$

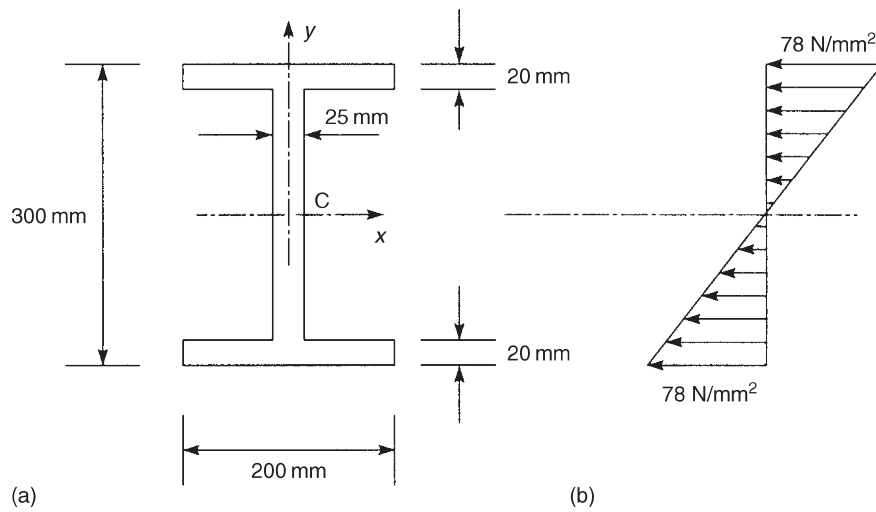


Fig. 16.6 Direct stress distribution in beam of Example 16.1.

The direct stress, therefore, varies linearly through the depth of the section from a value

$$-0.52 \times (+150) = -78 \text{ N/mm}^2 \text{ (compression)}$$

at the top of the beam to

$$-0.52 \times (-150) = +78 \text{ N/mm}^2 \text{ (tension)}$$

at the bottom as shown in Fig. 16.6(b).

Example 16.2

Now determine the distribution of direct stress in the beam of Example 16.1 if the bending moment is applied in a horizontal plane and in a clockwise sense about C_y when viewed in the direction yC .

In this case the beam will bend about the vertical y axis which therefore becomes the neutral axis of the section. Thus Eq. (16.9) becomes

$$\sigma_z = \frac{M}{I_{yy}} x \quad (\text{i})$$

where I_{yy} is the second moment of area of the beam section about the y axis. Again from Section 16.4

$$I_{yy} = 2 \times \frac{20 \times 200^3}{12} + \frac{260 \times 25^3}{12} = 27.0 \times 10^6 \text{ mm}^4$$

Hence, substituting for M and I_{yy} in Eq. (i)

$$\sigma_z = \frac{100 \times 10^6}{27.0 \times 10^6} x = 3.7x$$

We have not specified a sign convention for bending moments applied in a horizontal plane. However, a physical appreciation of the problem shows that the left-hand edges of the beam are in compression while the right-hand edges are in tension. Again the distribution is linear and varies from $3.7 \times (-100) = -370 \text{ N/mm}^2$ (compression) at the left-hand edges of each flange to $3.7 \times (+100) = +370 \text{ N/mm}^2$ (tension) at the right-hand edges.

We note that the maximum stresses in this example are very much greater than those in Example 16.1. This is due to the fact that the bulk of the material in the beam section is concentrated in the region of the neutral axis where the stresses are low. The use of an I-section in this manner would therefore be structurally inefficient.

Example 16.3

The beam section of Example 16.1 is subjected to a bending moment of 100 kN m applied in a plane parallel to the longitudinal axis of the beam but inclined at 30° to the left of vertical. The sense of the bending moment is clockwise when viewed from the left-hand edge of the beam section. Determine the distribution of direct stress.

The bending moment is first resolved into two components, M_x in a vertical plane and M_y in a horizontal plane. Equation (16.9) may then be written in two forms

$$\sigma_z = \frac{M_x}{I_{xx}}y \quad \sigma_z = \frac{M_y}{I_{yy}}x \quad (\text{i})$$

The separate distributions can then be determined and superimposed. A more direct method is to combine the two equations (i) to give the total direct stress at any point (x, y) in the section. Thus

$$\sigma_z = \frac{M_x}{I_{xx}}y + \frac{M_y}{I_{yy}}x \quad (\text{ii})$$

Now

$$\left. \begin{aligned} M_x &= 100 \cos 30^\circ = 86.6 \text{ kN m} \\ M_y &= 100 \sin 30^\circ = 50.0 \text{ kN m} \end{aligned} \right\} \quad (\text{iii})$$

M_x is, in this case, a positive bending moment producing tension in the upper half of the beam where y is positive. Also M_y produces tension in the left-hand half of the beam where x is negative; we shall therefore call M_y a negative bending moment. Substituting the values of M_x and M_y from Eq. (iii) but with the appropriate sign in Eq. (ii) together with the values of I_{xx} and I_{yy} from Examples 16.1 and 16.2 we obtain

$$\sigma_z = \frac{86.6 \times 10^6}{193.7 \times 10^6}y - \frac{50.0 \times 10^6}{27.0 \times 10^6}x \quad (\text{iv})$$

or

$$\sigma_z = 0.45y - 1.85x \quad (\text{v})$$

Equation (v) gives the value of direct stress at any point in the cross-section of the beam and may also be used to determine the distribution over any desired portion. Thus on

the upper edge of the top flange $y = +150$ mm, $100 \text{ mm} \geq x \geq -100$ mm, so that the direct stress varies linearly with x . At the top left-hand corner of the top flange

$$\sigma_z = 0.45 \times (+150) - 1.85 \times (-100) = +252.5 \text{ N/mm}^2 \text{ (tension)}$$

At the top right-hand corner

$$\sigma_z = 0.45 \times (+150) - 1.85 \times (+100) = -117.5 \text{ N/mm}^2 \text{ (compression)}$$

The distributions of direct stress over the outer edge of each flange and along the vertical axis of symmetry are shown in Fig. 16.7. Note that the neutral axis of the beam section does not in this case coincide with either the x or y axis, although it still passes through the centroid of the section. Its inclination, α , to the x axis, say, can be found by setting $\sigma_z = 0$ in Eq. (v). Then

$$0 = 0.45y - 1.85x$$

or

$$\frac{y}{x} = \frac{1.85}{0.45} = 4.11 = \tan \alpha$$

which gives

$$\alpha = 76.3^\circ$$

Note that α may be found in general terms from Eq. (ii) by again setting $\sigma_z = 0$. Hence

$$\frac{y}{x} = -\frac{M_y I_{xx}}{M_x I_{yy}} = \tan \alpha \quad (16.12)$$

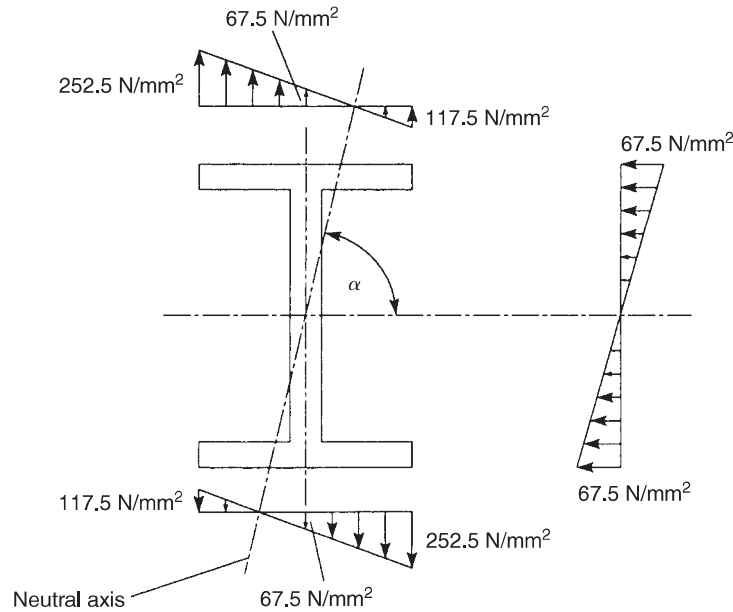


Fig. 16.7 Direct stress distribution in beam of Example 16.3.

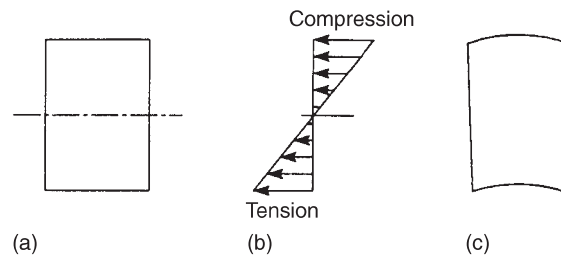


Fig. 16.8 Anticlastic bending of a beam section.

or

$$\tan \alpha = \frac{M_y I_{xx}}{M_x I_{yy}}$$

since y is positive and x is positive for a positive value of α . We shall define in a slightly different way in Section 16.2.4 for beams of unsymmetrical section.

16.1.3 Anticlastic bending

In the rectangular beam section shown in Fig. 16.8(a) the direct stress distribution due to a negative bending moment applied in a vertical plane varies from compression in the upper half of the beam to tension in the lower half (Fig. 16.8(b)). However, due to the Poisson effect the compressive stress produces a lateral elongation of the upper fibres of the beam section while the tensile stress produces a lateral contraction of the lower. The section does not therefore remain rectangular but distorts as shown in Fig. 16.8(c); the effect is known as *anticlastic bending*.

Anticlastic bending is of interest in the analysis of thin-walled box beams in which the cross-sections are maintained by stiffening ribs. The prevention of anticlastic distortion induces local variations in stress distributions in the webs and covers of the box beam and also in the stiffening ribs.

16.2 Unsymmetrical bending

We have shown that the value of direct stress at a point in the cross-section of a beam subjected to bending depends on the position of the point, the applied loading and the geometric properties of the cross-section. It follows that it is of no consequence whether or not the cross-section is open or closed. We therefore derive the theory for a beam of arbitrary cross-section and then discuss its application to thin-walled open and closed section beams subjected to bending moments.

The assumptions are identical to those made for symmetrical bending and are listed in Section 16.1.1. However, before we derive an expression for the direct stress distribution in a beam subjected to bending we shall establish sign conventions for moments, forces and displacements, investigate the effect of choice of section on the positive directions of these parameters and discuss the determination of the components of a bending moment applied in any longitudinal plane.

16.2.1 Sign conventions and notation

Forces, moments and displacements are referred to an arbitrary system of axes $Oxyz$, of which Oz is parallel to the longitudinal axis of the beam and Oxy are axes in the plane of the cross-section. We assign the symbols M , S , P , T and w to bending moment, shear force, axial or direct load, torque and distributed load intensity, respectively, with suffixes where appropriate to indicate sense or direction. Thus, M_x is a bending moment about the x axis, S_x is a shear force in the x direction and so on. Figure 16.9 shows positive directions and senses for the above loads and moments applied externally to a beam and also the positive directions of the components of displacement u , v and w of any point in the beam cross-section parallel to the x , y and z axes, respectively. A further condition defining the signs of the bending moments M_x and M_y is that they are positive when they induce tension in the positive xy quadrant of the beam cross-section.

If we refer internal forces and moments to that face of a section which is seen when viewed in the direction zO then, as shown in Fig. 16.10, positive internal forces and moments are in the same direction and sense as the externally applied loads whereas on the opposite face they form an opposing system. The former system, which we shall use, has the advantage that direct and shear loads are always positive in the positive directions of the appropriate axes whether they are internal loads or not. It must be realized, though, that internal stress resultants then become equivalent to externally applied forces and moments and are not in equilibrium with them.

16.2.2 Resolution of bending moments

A bending moment M applied in any longitudinal plane parallel to the z axis may be resolved into components M_x and M_y by the normal rules of vectors. However, a visual appreciation of the situation is often helpful. Referring to Fig. 16.11 we see that a bending moment M in a plane at an angle θ to Ox may have components of differing

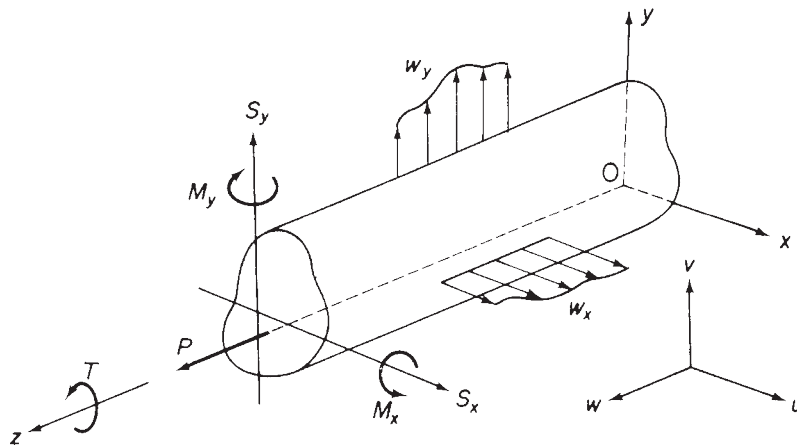


Fig. 16.9 Notation and sign convention for forces, moments and displacements.

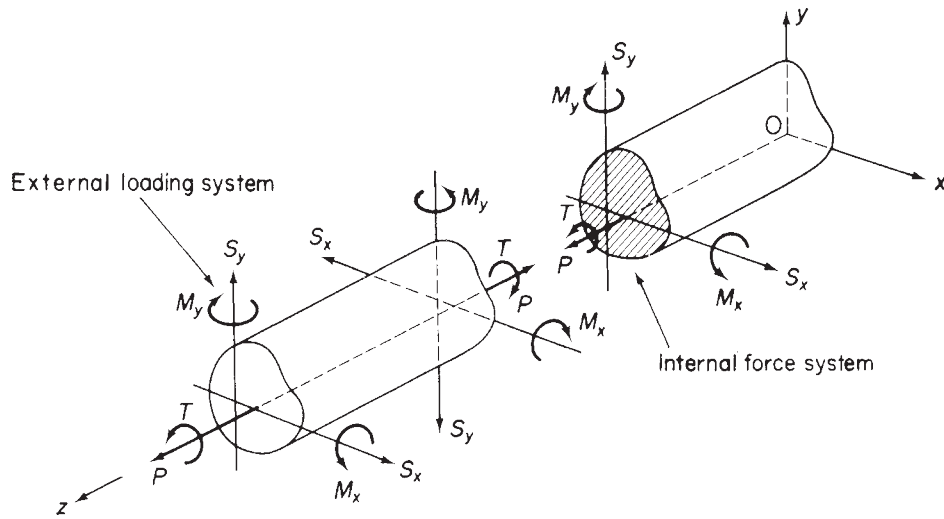


Fig. 16.10 Internal force system.

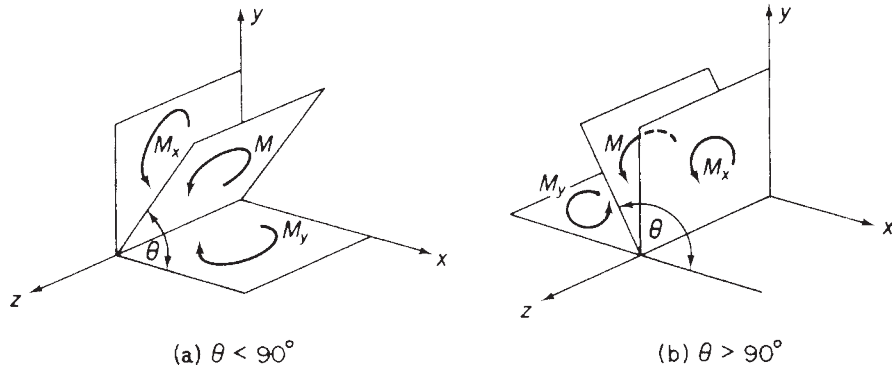


Fig. 16.11 Resolution of bending moments.

sign depending on the size of θ . In both cases, for the sense of M shown

$$M_x = M \sin \theta$$

$$M_y = M \cos \theta$$

which give, for $\theta < \pi/2$, M_x and M_y positive (Fig. 16.11(a)) and for $\theta > \pi/2$, M_x positive and M_y negative (Fig. 16.11(b)).

16.2.3 Direct stress distribution due to bending

Consider a beam having the arbitrary cross-section shown in Fig. 16.12(a). The beam supports bending moments M_x and M_y and bends about some axis in its cross-section which is therefore an axis of zero stress or a *neutral axis* (NA). Let us suppose that the

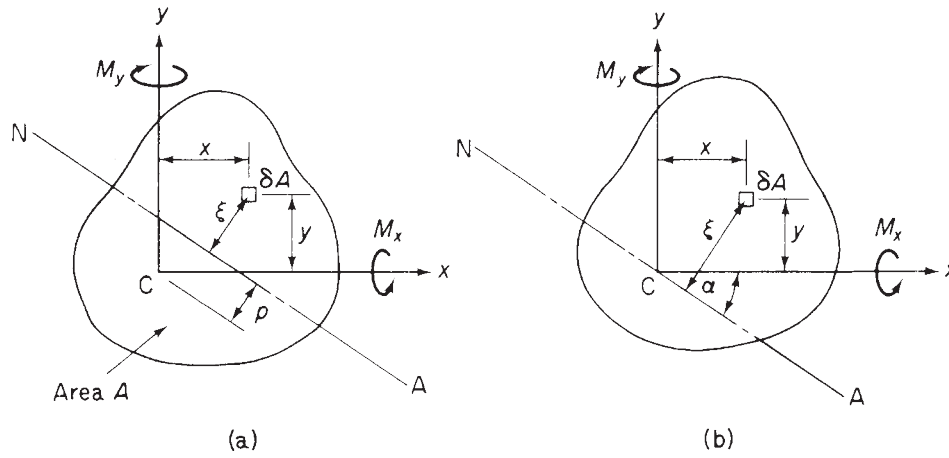


Fig. 16.12 Determination of neutral axis position and direct stress due to bending.

origin of axes coincides with the centroid C of the cross-section and that the neutral axis is a distance p from C . The direct stress σ_z on an element of area δA at a point (x, y) and a distance ξ from the neutral axis is, from the third of Eq. (1.42)

$$\sigma_z = E\varepsilon_z \quad (16.13)$$

If the beam is bent to a radius of curvature ρ about the neutral axis at this particular section then, since plane sections are assumed to remain plane after bending, and by a comparison with symmetrical bending theory

$$\varepsilon_z = \frac{\xi}{\rho}$$

Substituting for ε_z in Eq. (16.13) we have

$$\sigma_z = \frac{E\xi}{\rho} \quad (16.14)$$

The beam supports pure bending moments so that the resultant normal load on any section must be zero. Hence

$$\int_A \sigma_z dA = 0$$

Therefore, replacing σ_z in this equation from Eq. (16.14) and cancelling the constant E/ρ gives

$$\int_A \xi dA = 0$$

i.e. the first moment of area of the cross-section of the beam about the neutral axis is zero. It follows that the neutral axis passes through the centroid of the cross-section as shown in Fig. 16.12(b) which is the result we obtained for the case of symmetrical bending.

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Suppose that the inclination of the neutral axis to Cx is α (measured clockwise from Cx), then

$$\xi = x \sin \alpha + y \cos \alpha \quad (16.15)$$

and from Eq. (16.14)

$$\sigma_z = \frac{E}{\rho}(x \sin \alpha + y \cos \alpha) \quad (16.16)$$

The moment resultants of the internal direct stress distribution have the same sense as the applied moments M_x and M_y . Therefore

$$M_x = \int_A \sigma_z y \, dA, \quad M_y = \int_A \sigma_z x \, dA \quad (16.17)$$

Substituting for σ_z from Eq. (16.16) in (16.17) and defining the second moments of area of the section about the axes Cx, Cy as

$$I_{xx} = \int_A y^2 \, dA, \quad I_{yy} = \int_A x^2 \, dA, \quad I_{xy} = \int_A xy \, dA$$

gives

$$M_x = \frac{E \sin \alpha}{\rho} I_{xy} + \frac{E \cos \alpha}{\rho} I_{xx}, \quad M_y = \frac{E \sin \alpha}{\rho} I_{yy} + \frac{E \cos \alpha}{\rho} I_{xy}$$

or, in matrix form

$$\begin{Bmatrix} M_x \\ M_y \end{Bmatrix} = \frac{E}{\rho} \begin{bmatrix} I_{xy} & I_{xx} \\ I_{yy} & I_{xy} \end{bmatrix} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix}$$

from which

$$\frac{E}{\rho} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix} = \begin{bmatrix} I_{xy} & I_{xx} \\ I_{yy} & I_{xy} \end{bmatrix}^{-1} \begin{Bmatrix} M_x \\ M_y \end{Bmatrix}$$

i.e.

$$\frac{E}{\rho} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix} = \frac{1}{I_{xx}I_{yy} - I_{xy}^2} \begin{bmatrix} -I_{xy} & I_{xx} \\ I_{yy} & -I_{xy} \end{bmatrix} \begin{Bmatrix} M_x \\ M_y \end{Bmatrix}$$

so that, from Eq. (16.16)

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx}I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx}I_{yy} - I_{xy}^2} \right) y \quad (16.18)$$

Alternatively, Eq. (16.18) may be rearranged in the form

$$\sigma_z = \frac{M_x(I_{yy}y - I_{xy}x)}{I_{xx}I_{yy} - I_{xy}^2} + \frac{M_y(I_{xx}x - I_{xy}y)}{I_{xx}I_{yy} - I_{xy}^2} \quad (16.19)$$

From Eq. (16.19) it can be seen that if, say, $M_y = 0$ the moment M_x produces a stress which varies with both x and y ; similarly for M_y if $M_x = 0$.

In the case where the beam cross-section has *either* (or both) Cx or Cy as an axis of symmetry the product second moment of area I_{xy} is zero and Cx and Cy are *principal axes*. Equation (16.19) then reduces to

$$\sigma_z = \frac{M_x}{I_{xx}}y + \frac{M_y}{I_{yy}}x \quad (16.20)$$

Further, if either M_y or M_x is zero then

$$\sigma_z = \frac{M_x}{I_{xx}}y \quad \text{or} \quad \sigma_z = \frac{M_y}{I_{yy}}x \quad (16.21)$$

Equations (16.20) and (16.21) are those derived for the bending of beams having at least a singly symmetrical cross-section (see Section 16.1). It may also be noted that in Eq. (16.21) $\sigma_z = 0$ when, for the first equation, $y = 0$ and for the second equation when $x = 0$. Therefore, in symmetrical bending theory the x axis becomes the neutral axis when $M_y = 0$ and the y axis becomes the neutral axis when $M_x = 0$. Thus we see that the position of the neutral axis depends on the form of the applied loading as well as the geometrical properties of the cross-section.

There exists, in any unsymmetrical cross-section, a centroidal set of axes for which the product second moment of area is zero (see Ref. [1]). These axes are then principal axes and the direct stress distribution referred to these axes takes the simplified form of Eqs (16.20) or (16.21). It would therefore appear that the amount of computation can be reduced if these axes are used. This is not the case, however, unless the principal axes are obvious from inspection since the calculation of the position of the principal axes, the principal sectional properties and the coordinates of points at which the stresses are to be determined consumes a greater amount of time than direct use of Eqs (16.18) or (16.19) for an arbitrary, but convenient set of centroidal axes.

16.2.4 Position of the neutral axis

The neutral axis always passes through the centroid of area of a beam's cross-section but its inclination α (see Fig. 16.12(b)) to the x axis depends on the form of the applied loading and the geometrical properties of the beam's cross-section.

At all points on the neutral axis the direct stress is zero. Therefore, from Eq. (16.18)

$$0 = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x_{NA} + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y_{NA}$$

where x_{NA} and y_{NA} are the coordinates of any point on the neutral axis. Hence

$$\frac{y_{NA}}{x_{NA}} = - \frac{M_y I_{xx} - M_x I_{xy}}{M_x I_{yy} - M_y I_{xy}}$$

or, referring to Fig. 16.12(b) and noting that when α is positive x_{NA} and y_{NA} are of opposite sign

$$\tan \alpha = \frac{M_y I_{xx} - M_x I_{xy}}{M_x I_{yy} - M_y I_{xy}} \quad (16.22)$$

Example 16.4

A beam having the cross-section shown in Fig. 16.13 is subjected to a bending moment of 1500 N m in a vertical plane. Calculate the maximum direct stress due to bending stating the point at which it acts.

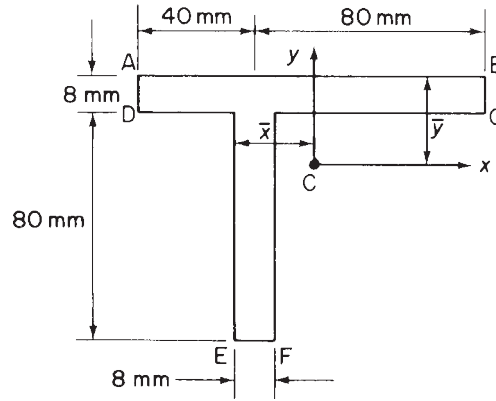


Fig. 16.13 Cross-section of beam in Example 16.4.

The position of the centroid of the section may be found by taking moments of areas about some convenient point. Thus

$$(120 \times 8 + 80 \times 8)\bar{y} = 120 \times 8 \times 4 + 80 \times 8 \times 48$$

giving

$$\bar{y} = 21.6 \text{ mm}$$

and

$$(120 \times 8 + 80 \times 8)\bar{x} = 80 \times 8 \times 4 + 120 \times 8 \times 24$$

giving

$$\bar{x} = 16 \text{ mm}$$

The next step is to calculate the section properties referred to axes Cxy (see Section 16.4)

$$\begin{aligned} I_{xx} &= \frac{120 \times (8)^3}{12} + 120 \times 8 \times (17.6)^2 + \frac{8 \times (80)^3}{12} + 80 \times 8 \times (26.4)^2 \\ &= 1.09 \times 10^6 \text{ mm}^4 \end{aligned}$$

$$\begin{aligned} I_{yy} &= \frac{8 \times (120)^3}{12} + 120 \times 8 \times (8)^2 + \frac{80 \times (8)^3}{12} + 80 \times 8 \times (12)^2 \\ &= 1.31 \times 10^6 \text{ mm}^4 \end{aligned}$$

$$\begin{aligned} I_{xy} &= 120 \times 8 \times 8 \times 17.6 + 80 \times 8 \times (-12) \times (-26.4) \\ &= 0.34 \times 10^6 \text{ mm}^4 \end{aligned}$$

Since $M_x = 1500 \text{ N m}$ and $M_y = 0$ we have, from Eq. (16.19)

$$\sigma_z = 1.5y - 0.39x \quad (i)$$

in which the units are N and mm.

By inspection of Eq. (i) we see that σ_x will be a maximum at F where $x = -8 \text{ mm}$, $y = -66.4 \text{ mm}$. Thus

$$\sigma_{z,\max} = -96 \text{ N/mm}^2 \text{ (compressive)}$$

In some cases the maximum value cannot be obtained by inspection so that values of σ_z at several points must be calculated.

16.2.5 Load intensity, shear force and bending moment relationships, general case

Consider an element of length δz of a beam of unsymmetrical cross-section subjected to shear forces, bending moments and a distributed load of varying intensity, all in the yz plane as shown in Fig. 16.14. The forces and moments are positive in accordance with the sign convention previously adopted. Over the length of the element we may assume that the intensity of the distributed load is constant. Therefore, for equilibrium of the element in the y direction

$$\left(S_y + \frac{\partial S_y}{\partial z} \delta z \right) + w_y \delta z - S_y = 0$$

from which

$$w_y = -\frac{\partial S_y}{\partial z}$$

Taking moments about A we have

$$\left(M_x + \frac{\partial M_x}{\partial z} \delta z \right) - \left(S_y + \frac{\partial S_y}{\partial z} \delta z \right) \delta z - w_y \frac{(\delta z)^2}{2} - M_x = 0$$

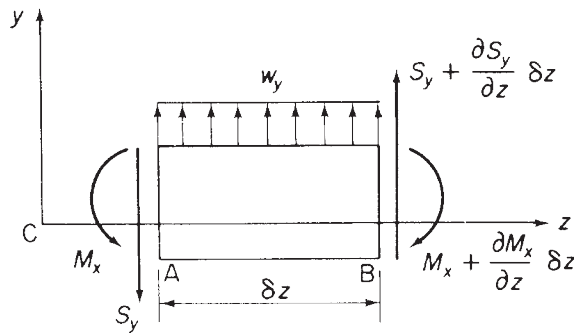


Fig. 16.14 Equilibrium of beam element supporting a general force system in the yz plane.